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Enhanced phosphate removal from piggery wastewater using electrocoagulation and RSM optimization for eutrophication control

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Abstract. Phosphorus-rich wastewater from pig farming poses a major threat to aquatic ecosystems due to its contribution to eutrophication. This study investigated the removal of phosphate from actual piggery wastewater using electrocoagulation, aiming to optimize operational parameters for maximum treatment efficiency. A Central Composite Design (CCD) under Response Surface Methodology (RSM) was employed to evaluate the combined effects of pH, current intensity, and reaction time on phosphate removal. Experimental results revealed that pH played the most crucial role, significantly influencing the hydrolysis of aluminum, which is essential for effective phosphate adsorption and co-precipitation. Current intensity and reaction time also supported the process but were less dominant. The quadratic regression model showed a strong fit ($F = 112.79$, $p < 0.0001$) with excellent predictive capability. Optimal conditions were determined at pH 8, a current of approximately 3 A, and a reaction time of 25 minutes, achieving a predicted phosphate removal efficiency of 96.47%, which was subsequently validated by repeated experiments. Further tests indicated that maintaining reaction times between 20 minutes ensured near-complete phosphate removal while minimizing energy consumption. This research provides a solid scientific basis for implementing electrocoagulation as a cost-effective and sustainable solution to manage nutrient-rich wastewater from intensive pig farming in Vietnam.

1. Introduction

The pig farming sector in Vietnam is experiencing robust growth and represents a substantial proportion of the country's agricultural output. However, this rapid expansion has led to significant challenges related to environmental pollution, primarily due to the large volumes of wastewater generated [1]. This wastewater typically contains high concentrations of organic matter, ammoniacal nitrogen, phosphorus, and even heavy metals. Notably, a large share of wastewater from pig farms still goes untreated, posing a serious threat to surface and groundwater quality. Traditional treatment approaches, including biological (anaerobic and aerobic) and physico-chemical methods (such as chemical precipitation and neutralization),



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currently address only about 40% of the actual demand [1, 2]. These systems often reveal considerable shortcomings, including high operational costs, unstable treatment efficiencies, difficulties in scaling up, and the generation of secondary sludge [3]. For instance, small-scale biogas digesters, which are widely adopted by household farms, are inadequate to cope with the increased waste loads, frequently resulting in effluents that fail to meet discharge standards. This underscores an urgent need for supplementary or alternative solutions [2].

Against this backdrop, electrochemical technologies have emerged as a promising avenue for treating wastewater streams with heavy pollution loads. A particularly advantageous feature of piggery wastewater is its inherently high electrical conductivity approximately 6 mS/cm according to several studies which facilitates electrochemical processes without the need for external electrolytes. Among these technologies, electrocoagulation (EC) stands out for its ability to remove a wide spectrum of pollutants [4]. EC relies on sacrificial metal electrodes, commonly aluminum or iron, to generate metal ions in situ, which subsequently hydrolyze and form flocs that capture and eliminate contaminants from the liquid phase [2, 5]. Numerous studies have demonstrated that EC is highly effective in treating piggery wastewater, achieving substantial removal of organic compounds, phosphorus, and heavy metals. Under optimal conditions using aluminum electrodes, the total phosphorus (TP) removal efficiency can reach around 96%, along with significant reductions in turbidity and suspended solids, making it a promising solution to tackle nutrient-induced eutrophication originating from livestock wastewater [6]. Although electrocoagulation (EC) has been extensively investigated for phosphate removal, most previous studies were conducted under controlled laboratory conditions using synthetic wastewater with moderate pollutant loads and often required the addition of supporting electrolytes or operation under highly acidic environments. For instance, Attour et al. reported a maximum removal efficiency of approximately 90% from a synthetic solution (100 mg/L) at an initial pH of 3, where the dominant mechanism was aluminum phosphate precipitation [5]. Similarly, Dura and Breslin achieved complete phosphate removal within 30–60 minutes using Al–Mg alloy anodes at a current density of 11 mA/cm², but the system required 4.2 mM NaCl to enhance conductivity, and efficiency declined at higher initial concentrations [7, 8]. In contrast, the present study focuses on the treatment of real piggery wastewater in Vietnam, which is characterized by exceptionally high organic and nutrient loads and naturally elevated electrical conductivity ($\approx 1.3\text{--}6.0$ mS/cm), eliminating the need for additional electrolytes [2]. Leveraging these unique physicochemical properties, this work demonstrates that phosphate removal is highly effective due to the formation of porous Al(OH)₃ flocs, which enhance both adsorption and co-precipitation processes. Consequently, the proposed approach provides a cost-effective, sustainable, and scalable solution for mitigating eutrophication risks and promoting resource recovery in regions with intensive pig farming [9–12].

Nevertheless, the performance of the electrocoagulation process is closely tied to multiple operational parameters, such as pH, current density, electrolysis time, electrode spacing, and electrolyte concentration. If these parameters are not carefully optimized and controlled, treatment efficiency may be compromised, and energy costs could rise, diminishing the economic feasibility of the process [13]. Consequently, there is a pressing need to optimize these operational factors simultaneously to achieve maximum phosphorus removal at minimal cost. In this study, Response Surface Methodology (RSM) is employed as a powerful statistical tool to design experiments, assess the combined effects of multiple variables, and identify optimal operating conditions with a minimal number of experimental runs [14]. By determining the best values for key parameters such as initial pH, current density, and reaction time, this research aims to

enhance phosphorus removal efficiency, reduce operational costs, and lay a scientific foundation for the practical application of electrochemical technology in treating piggery wastewater in Vietnam in a sustainable manner.

2. Materials and research methods

2.1 Research Materials

The wastewater used in this study was collected from the effluent stream following the anaerobic biogas treatment tank of Phat Loc Phat pig farm, located at No. 7, Hamlet 8B, Loc Hoa Commune, Loc Ninh District, Binh Phuoc Province, Vietnam. This source represents typical wastewater generated from large-scale pig farming operations, characterized by a notably high phosphate content that plays a significant role in triggering eutrophication if left untreated. Initial analyses revealed that the wastewater had a pH of approximately 6.3, an electrical conductivity of 1340 $\mu\text{S}/\text{cm}$, and a particularly high phosphate concentration measured at $55.53 \pm 0.51 \text{ mg/L}$, indicating substantial nutrient pollution.

Table 1. Experimental design for predicting phosphate removal efficiency.

Run	pH	Current intensity (A)	Reaction time (minutes)
1	4	2	35
2	8	2	5
3	8	1	20
4	4	3	20
5	8	2	35
6	6	2	20
7	4	1	20
8	6	3	35
9	6	2	20
10	4	2	5
11	6	2	20
12	8	3	20
13	6	1	35
14	6	1	5
15	6	3	5

The experiments were carried out in an electrochemical system comprising a rectangular acrylic (mica) reaction tank with dimensions of 15 cm (length) $\times$ 15 cm (width) $\times$ 16.5 cm (height).

In this setup, aluminum was selected as the electrode material, arranged with two plates for each pole. Each plate measured $100 \times 170 \times 5$ mm and was installed in parallel, with adjustable spacing to accommodate different experimental conditions. A direct current (DC) power supply delivered electricity directly to the electrodes, allowing precise adjustment and monitoring of both voltage and current intensity displayed on the device. This design ensured stable electrochemical reactions while providing flexibility and control over the operational parameters throughout the experiments.

2.2 Research Methodology and Analysis.

The wastewater The experimental design (DoE) employed in this study aimed to optimize the removal of phosphate from wastewater using the Response Surface Methodology (RSM). Three key operational variables were selected for investigation: pH, current intensity (A), and reaction time (minutes) [13]. These factors were organized and evaluated under a Central Composite Design (CCD) framework, which enabled the prediction and determination of the optimal values for each parameter. Each variable was examined at three levels, corresponding to low (−1), medium (0), and high (+1) settings within the CCD structure [15]. Based on these selected factors, the Design-Expert software generated a set of 15 experimental runs to comprehensively explore the experimental space. Details of these combinations are presented in Table 1. This approach allowed for a robust assessment of phosphate removal efficiency through the electrochemical system under various operating conditions.

3. Results And Discussion

3.1 Effect of operational factors on phosphate removal efficiency

The experimental results are presented in Figure 1, illustrating the effects of pH, current intensity, and reaction time on the efficiency of phosphate removal from piggery wastewater using the electrocoagulation process. Figure 1A shows the relationship between pH and current intensity. It is evident that the removal efficiency increased substantially when the pH was adjusted from near neutral to mildly alkaline conditions. Specifically, at a pH of approximately 8 combined with a current intensity of 3 A, the phosphate removal efficiency reached its highest value of 96.5%. In contrast, maintaining the same current but lowering the pH to 4 resulted in an efficiency of only 41.5%, nearly half, highlighting the critical role of pH in regulating the hydrolysis of aluminum and the formation of $\text{Al}(\text{OH})_3$ complexes key mechanisms for phosphate adsorption and co-precipitation. Even at a lower current of 1 A, the efficiency still achieved 80.8% at pH 8, compared to merely 23.9% at pH 4 after 35 minutes, further underscoring the dominant influence of pH relative to current intensity alone.

The combined effect of pH and reaction time is depicted in Figure 1B. When pH was kept low, around 4, extending the reaction time from 5 to 35 minutes only increased the removal efficiency slightly, ranging from 9.9% to 25.4%. This indicates that simply prolonging contact time is insufficient to drive the formation and stabilization of aluminum hydroxide flocs if pH conditions are not favorable. Conversely, at pH 8, even a short reaction time of 5 minutes yielded an efficiency of 51.1%, which further rose to 64.7% at 20 minutes, demonstrating that an alkaline environment effectively facilitates the hydrolysis of aluminum, leading to the creation of porous $\text{Al}(\text{OH})_3$ networks that enhance phosphate capture. Notably, at pH 8, with a current of 2 A and a reaction time of 20 minutes, the efficiency reached 64.7%, reaffirming the pivotal role of pH in electrocoagulation.

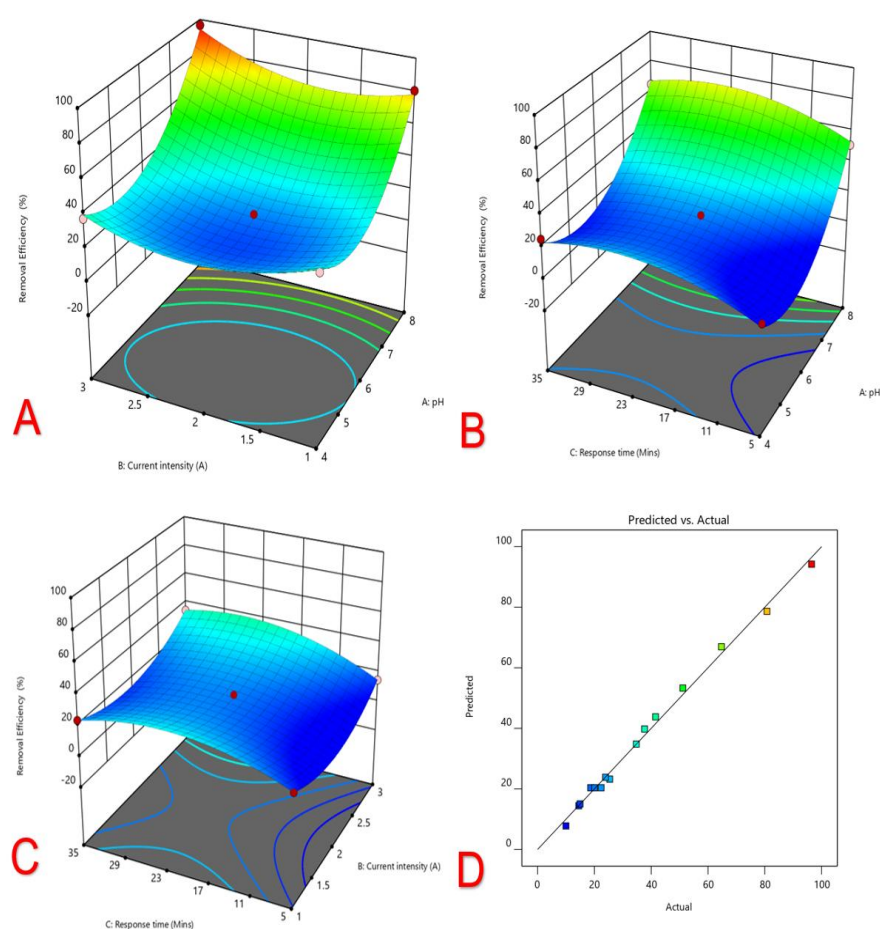


Figure 1. Optimization of phosphate removal efficiency under different conditions: (A) pH and current intensity; (B) pH and reaction time; (C) current intensity and reaction time; (D) correlation between experimental results and predicted values.

Figure 1C illustrates the interaction between current intensity and reaction time. Although increasing both parameters slightly improved removal efficiency, the effect was far less pronounced in the absence of pH adjustment. At pH 6, raising the current from 1 A to 3 A and extending the time from 5 to 35 minutes only increased efficiency from 14.9% to a maximum of 37.7%, which is considerably lower than under optimal pH conditions. This suggests that while higher current generates more Al^{3+} ions and longer reaction times allow more complete hydrolysis and floc formation, neither factor alone can compensate for the lack of an alkaline environment necessary for stable hydroxide formation and effective phosphate precipitation.

The ANOVA analysis further confirmed the robustness of the model. The quadratic regression model showed a highly significant fit ($F = 112.79$; $p < 0.0001$), with pH identified as the most influential factor ($F = 436.39$), followed by reaction time ($F = 46.46$) and current ($F = 7.33$). The squared terms and the interaction between pH and current were also statistically significant. The high coefficients of determination ($R^2 = 0.9951$, Predicted $R^2 = 0.9319$) and an Adeq Precision of 35.04 demonstrated excellent predictive capability and model reliability for optimizing phosphate removal.

Based on these results, the regression equation in coded form expressing the dependence of removal efficiency on the three key operational factors was established as:

$$\text{Removal Efficiency(\%)} = 274.2 - 82.71A - 1.1B + 2.4C + 2.44AB - 0.015AC + 0.17BC + 7.43A^2 + 13.9B^2 - 0.05C^2$$

where A is pH, B is current intensity (A), and C is reaction time (minutes).

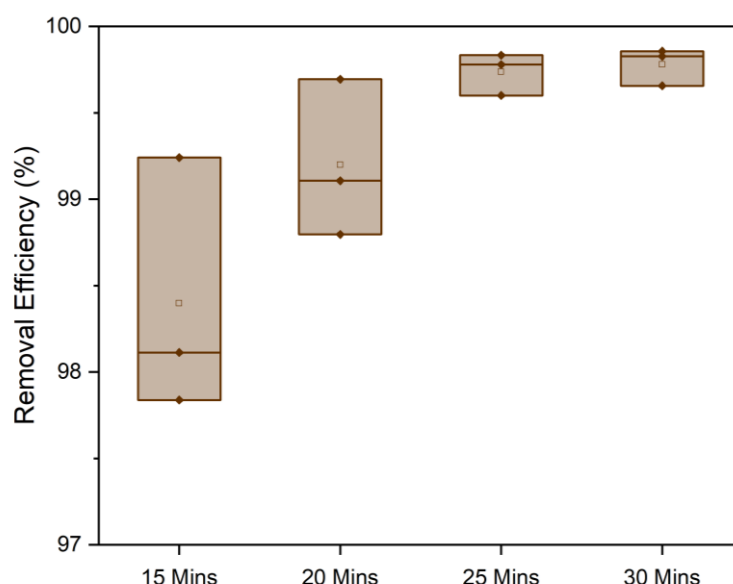


Figure 2. Phosphate removal efficiencies across these different reaction times.

Optimization results determined that the optimal operational conditions were pH 8, a current of approximately 3 A, and a reaction time of 25 minutes, with a desirability of 100%. Under these conditions, the predicted phosphate removal efficiency was 96.47%, confirming the effectiveness of the optimized electrocoagulation process for treating piggery wastewater. When compared to the findings of previous studies, this research demonstrates notable advancements in phosphate removal efficiency and operational feasibility. Kuokkanen et al. achieved a maximum phosphate removal efficiency of 96% when treating synthetic wastewater (30 mg/L) using a hybrid Al/Fe electrode system at pH 5 with the addition of 1 g/L NaCl as a supporting electrolyte [16]. In the same study, treatment of real wastewater resulted in 93% removal for dairy effluents and 79% for mining wastewater, requiring longer reaction times of 30–60 minutes. Similarly, Attour et al. investigated phosphate removal from a synthetic solution (100 mg/L) using aluminum electrodes and reported a maximum efficiency of approximately 90% under strongly acidic conditions (pH 3), where the dominant mechanism was aluminum phosphate precipitation [17].

3.2 Evaluation of the effect of reaction time after process optimization

Following the optimization process, the optimal operating conditions were identified, including a mildly alkaline pH, an appropriate current intensity, and a suitable reaction time. Prior investigations had already demonstrated that pH plays the most critical role in phosphate removal efficiency, while current intensity and reaction time act as supportive factors that ensure the effective hydrolysis of aluminum and the formation of sufficiently large $\text{Al}(\text{OH})_3$ complexes capable of adsorbing and co-precipitating phosphate from the aqueous phase. After establishing these optimal conditions, additional experiments were conducted to validate the results and fine-

tune the reaction time to achieve high removal efficiency while also optimizing operational costs. Four reaction times 15, 20, 25, and 30 minutes were selected for this assessment.

Specifically, at 15 minutes, the removal efficiency ranged from 97.8% to 99.2%, with an average of approximately 98.4%, although the relatively high dispersion, as illustrated by the elevated boxplot, indicated that the coagulation process was nearing completion but some fluctuations remained. Increasing the reaction time to 20 minutes resulted in a higher average efficiency of around 99.2%, accompanied by a markedly reduced variance, suggesting that hydrolysis and floc formation were nearly complete. Most notably, at 25 and 30 minutes, phosphate removal efficiencies consistently exceeded 99.6%, with minimal differences across replicates, demonstrating that the system had reached a stable equilibrium, leaving only negligible residual phosphate in the treated water. Tian et al. (2017) reported 98% removal within 15 minutes using a reverse-electric field air-cathode electrocoagulation (REAEC) system at a low current density (1 mA/cm^2), but their setup required a more complex three-electrode configuration and an external air cathode [18]. Dura and Breslin (2019) achieved complete phosphate removal within 30–60 minutes using Al–Mg alloy anodes at a higher current density (11 mA/cm^2) and required 4.2 mM NaCl supplementation, with efficiency decreasing at higher initial concentrations [7]. Similarly, Kuokkanen et al. (2014) obtained 96% removal for synthetic wastewater and 93% for dairy effluents, but the process required 1 g/L NaCl and longer treatment times (30–60 minutes). Attour et al. (2014) reported about 90% removal at a highly acidic pH 3, dominated by aluminum phosphate precipitation. In contrast, the present study achieved >99.6% removal within 25 minutes under mildly alkaline conditions ($\text{pH} \approx 8$) without external electrolytes, owing to the naturally high conductivity (1.34 mS/cm) of piggery wastewater. These results highlight the superior efficiency, shorter treatment times, and greater scalability of the proposed approach for nutrient-rich livestock wastewater treatment.

These findings not only validate the predictive capability of the optimization model but also highlight the most economically viable reaction window. A reaction time between 20 and 25 minutes appears sufficient to achieve near-maximal phosphate removal, while extending beyond 30 minutes offers little additional benefit. This has important economic implications, as selecting an optimal reaction time within this range ensures nearly complete phosphate removal while minimizing energy consumption, thereby enhancing the feasibility of scaling up the process for industrial applications

4. Conclusion

This study demonstrated that electrocoagulation is a highly effective and promising approach for removing phosphate from piggery wastewater, which typically contains high nutrient loads and poses significant environmental risks. By applying Response Surface Methodology (RSM) with a Central Composite Design (CCD), the combined effects of pH, current intensity, and reaction time on phosphate removal efficiency were systematically explored and optimized. Under optimal conditions pH 8, current intensity of approximately 3 A, and a reaction time of 25 minutes the phosphate removal efficiency was predicted to reach 96.47%, which was subsequently confirmed through additional validation experiments. Further assessments showed that maintaining the reaction time within 20 minutes was sufficient to achieve near-maximal removal, providing an effective balance between treatment performance and operational costs. Overall, these findings provide a robust scientific basis for the practical implementation of electrocoagulation technology in treating piggery wastewater in Vietnam. By achieving high phosphate removal efficiencies under optimized conditions, this approach not only mitigates the risk of eutrophication but also

offers a cost-effective and sustainable solution for managing wastewater in intensive livestock operations.

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